

Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)

PS ID: SIH26164*

Theme: Blockchain & Cybersecurity

Category: Software

Team ID: _____

Team Name: _____

CipherScope — the tool behind the pitch

A one-command scanner that discovers every piece of encryption in a codebase and tells you which parts will break when quantum computers arrive — and exactly what to replace them with.

Live demo (click):

cipherscope-dashboard-production.up.railway.app

***check the exact PS number shown on sih.gov.in → Problem Statements (site shows SIH26163/26164)**

CipherScope — Cryptographic Discovery & Quantum-Risk Scanner

Proposed Solution

- **Scan:** reads every source file + certificate + config in a repo (Python, Java, Go, C, JS, YAML, Dockerfile, sshd...) and lists every crypto artefact.
- **Assess:** flags **weak-today** vs **quantum-vulnerable**, rates severity (Critical→Low).
- **Time-check:** applies **Mosca's theorem** — data lifetime + migration time vs quantum horizon.
- **Fix:** maps each finding to a **NIST PQC replacement** (ML-KEM, ML-DSA...).
- **Show:** interactive GUI + raw JSON/exportable report.

How it addresses the PS

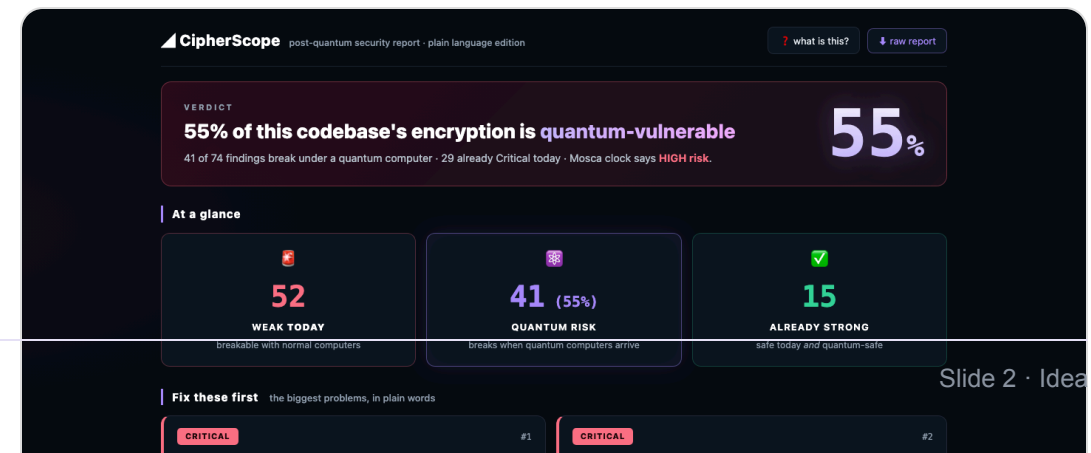
- Exactly the 4 asks: catalogue artefacts → quantum risk → Mosca classification → PQC alternatives, delivered through an interactive GUI.

Innovation & uniqueness

- **Three detection engines** (regex + Python AST parse + real X.509 parsing) = high accuracy.
- **Real Mosca calculation**, not just a label.
- **Plain-language dashboard** — non-experts (boards/CISOs) can read it in 10 seconds.

- **Live demo today:** 20 files → **74 findings**.

55% quantum-vulnerable · Mosca HIGH.



TECHNICAL APPROACH

How it works

Layer	What it does
1. Regex engine	<code>.py .java .js .go .c .cpp .yaml .sh</code> + <code>Dockerfile / sshd_config / nginx.conf</code> — catches algorithm names, key sizes, TLS settings, cipher lists.
2. Python AST engine	Parses code structure: <code>hashlib.*</code> , <code>Crypto.Cipher.*</code> , <code>ssl contexts</code> , <code>rsa/ec keys</code> — pattern-proof detection.
3. Certificate parsing	Reads real <code>.crt/.key/.pem</code> : key size, expiry, self-signed — found an expired RSA-1024 cert in the demo.
4. Mosca engine	<code>data_lifetime + migration_years > quantum_horizon</code> → HIGH/MED/LOW risk.
5. PQC mapper	<code>RSA→ML-KEM-768 · ECDSA→ML-DSA-65 · DES→AES-256-GCM · SHA-1→SHA-256</code> (NIST FIPS 203/204/205).
6. GUI	React + Vite dashboard: verdict, heatmap, filters, per-finding recommendations; served via Docker/nginx on Railway.

Verified pipeline

`python3 scanner.py demo-repo -o findings.json --data-lifetime 10 --migration-years 6 --quantum-horizon 15` → **20 files scanned · 74 findings (29 Critical) · 41/74 quantum-vulnerable · <1 second.**

FEASIBILITY & VIABILITY

Feasible today, viable tomorrow

Why it's feasible

- **No external calls** — works fully offline; scans any repo in seconds.
- **Cheap stack**: Python stdlib + cryptography lib + React; deployed free on Railway (Docker/nginx).
- **Already working**: full pipeline verified end-to-end with live URL.
- **Openable datasets** (per PS): GitHub repos, OpenSSL codebases.

Challenges & risks → strategy

- **False negatives** (obfuscated/embedded crypto) → layered engines + roadmap for binary/container scanning (ELF strings, SBOM/CBOM parse).
- **Business-criticality tagging** → per-service config file (who owns what data, retention years).
- **Scalability** → parallel file walker; works on monorepos.
- **Data freshness** → `npm run scan` re-runs scanner in one command; report export included.

Viability

Directly serves the **NIST 2030–2035 PQC migration mandate**. Every bank, ministry and enterprise doing the transition needs an inventory first — CipherScope is that inventory, in seconds.

IMPACT & BENEFITS

Why this matters (and for whom)

Target audience

- **Government ministries & PSUs** owning citizen data (Aadhaar-era systems, DigiLocker, UPI).
- **CISOs / security teams** starting PQC migration.
- **SOCs & auditors** — currently manual crypto audits.
- **Developers** — see exactly which lines to change.

Benefits

- **Speed:** 3-month manual audit → **1-minute scan**.
- **Prioritisation:** Critical-first remediation, so budgets go where risk is.
- **Harvest-now protection:** stops “steal encrypted data today, decrypt with quantum later”.
- **Compliance:** maps to NIST PQC standards & Mosca-based timelines — board-ready language.

India angle: as Bharat migrates digital public infrastructure to post-quantum security, every asset custodian needs a tool like this first. CipherScope is that first step.

RESEARCH & REFERENCES

Sources & links

- **NIST FIPS 203** (ML-KEM), **FIPS 204** (ML-DSA), **FIPS 205** (SLH-DSA) — post-quantum standards.
- **NIST IR 8547** — transition to post-quantum cryptography.
- **Mosca, M. (2015)** — “Cybersecurity in an era with quantum computers: will we be ready?” (IEEE Security & Privacy).
- **OWASP CycloneDX CBOM** — Cryptographic Bill of Materials standard (report-format roadmap).
- **Live demo:** <https://cipherscope-dashboard-production.up.railway.app>
- **Project docs:** scanner.py (514 lines), demo-repo (6 services), dashboard (React/Vite) — in submission repo.

One-liner to remember

“Quantum is coming. CipherScope finds every lock it will break — and hands you the new key.”